

Exercise 1: Configuring a Basic Spring Application

```
src > main > java > com > library > J MainApp.java > MainApp

1  package com.library;
2
3  import org.springframework.context.ApplicationContext;
4  import org.springframework.context.support.ClassPathXmlApplicationContext;
5
6  import com.library.service.BookService;
7
8  public class MainApp {
9
10     Run | Debug
11     public static void main(String[] args) {
12
13         ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml");
14
15         BookService service = context.getBean("bookService", BookService.class);
16
17         service.issueBook();
18     }
19 }
```

```
pom.xml > project
1  <?xml version="1.0" encoding="UTF-8"?>
2  <project xmlns="http://maven.apache.org/POM/4.0.0"
3  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
5  http://maven.apache.org/xsd/maven-4.0.0.xsd">
6
7      <modelVersion>4.0.0</modelVersion>
8
9      <groupId>com.library</groupId>
10     <artifactId>LibraryManagement</artifactId>
11     <version>1.0-SNAPSHOT</version>
12
13     <properties>
14         <maven.compiler.source>17</maven.compiler.source>
15         <maven.compiler.target>17</maven.compiler.target>
16     </properties>
17
18     <dependencies>
19         <!-- Spring Core -->
20         <dependency>
21             <groupId>org.springframework</groupId>
22             <artifactId>spring-context</artifactId>
23             <version>5.3.39</version>
24         </dependency>
25     </dependencies>
26
27     <build>
28         <plugins>
29             <plugin>
30                 <groupId>org.codehaus.mojo</groupId>
31                 <artifactId>exec-maven-plugin</artifactId>
32                 <version>3.1.0</version>
33             </plugin>
34         </plugins>
35     </build>
36
37 </project>
```

```
src > main > resources > applicationContext.xml > beans
1  <?xml version="1.0" encoding="UTF-8"?>
2
3  <beans xmlns="http://www.springframework.org/schema/beans"
4      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5
6      xsi:schemaLocation="
7          http://www.springframework.org/schema/beans
8          https://www.springframework.org/schema/beans/spring-beans.xsd">
9
10     <bean id="bookRepository"
11         class="com.library.repository.BookRepository"/>
12
13     <bean id="bookService"
14         class="com.library.service.BookService">
15
16         <property name="bookRepository"
17             ref="bookRepository"/>
18
19     </bean>
20
21 </beans>
```

```
src > main > java > com > library > repository > BookRepository.java > BookRepository
1  package com.library.repository;
2
3  public class BookRepository {
4
5      public void displayBooks() {
6          System.out.println(x: "Fetching books from repository...");
7      }
8
9  }
```

```
src > main > java > com > library > service > BookService.java > BookService
1  package com.library.service;
2
3  import com.library.repository.BookRepository;
4
5  public class BookService {
6
7      private BookRepository bookRepository;
8
9      public void setBookRepository(BookRepository bookRepository) {
10         this.bookRepository = bookRepository;
11     }
12
13     public void issueBook() {
14         System.out.println(x: "Book Service Called");
15         bookRepository.displayBooks();
16     }
17
18 }
```

```
Book Service Called
Fetching books from repository...
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.172 s
[INFO] Finished at: 2026-06-29T23:59:24+05:30
[INFO] -----
```

Exercise 2: Implementing Dependency Injection

```
src > main > java > com > library > MainApp.java > MainApp
1 package com.library;
2
3 import org.springframework.context.ApplicationContext;
4 import org.springframework.context.support.ClassPathXmlApplicationContext;
5
6 import com.library.service.BookService;
7
8 public class MainApp {
9
10     Run | Debug
11     public static void main(String[] args) {
12
13         ApplicationContext context =
14             new ClassPathXmlApplicationContext("applicationContext.xml");
15
16         BookService service =
17             context.getBean("bookService", BookService.class);
18
19         service.issueBook();
20
21         ((ClassPathXmlApplicationContext) context).close();
22     }
23 }

src > main > java > com > library > repository > BookRepository.java > BookRepository > displayBooks()
1 package com.library.repository;
2
3 public class BookRepository {
4
5     public void displayBooks() {
6         System.out.println(x: "BookRepository: Fetching books from the database...");
7     }
8 }
```

src > main > java > com > library > service > BookService.java > ...

```
1 package com.library.service;
2
3 import com.library.repository.BookRepository;
4
5 public class BookService {
6
7     private BookRepository bookRepository;
8
9     // Setter Injection
10    public void setBookRepository(BookRepository bookRepository) {
11        this.bookRepository = bookRepository;
12    }
13
14    public void issueBook() {
15        System.out.println(x: "BookService: Issuing a book...");
16        bookRepository.displayBooks();
17    }
18 }
```

src > main > resources > applicationContext.xml > beans

```
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5       xsi:schemaLocation="
6           http://www.springframework.org/schema/beans
7           https://www.springframework.org/schema/beans/spring-beans.xsd">
8
9     <!-- Repository Bean -->
10    <bean id="bookRepository"
11         class="com.library.repository.BookRepository"/>
12
13    <!-- Service Bean -->
14    <bean id="bookService"
15         class="com.library.service.BookService">
16
17        <!-- Dependency Injection -->
18        <property name="bookRepository" ref="bookRepository"/>
19
20    </bean>
21
22 </beans>
```

BookService: Issuing a book...

BookRepository: Fetching books from the database...

```
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.225 s
[INFO] Finished at: 2026-06-30T00:05:54+05:30
[INFO] -----
```

Exercise 4: Creating and Configuring a Maven Project

```
pom.xml > @project
1 <project xmlns="http://maven.apache.org/POM/4.0.0"
2   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
4     http://maven.apache.org/xsd/maven-4.0.0.xsd">
5
6   <modelVersion>4.0.0</modelVersion>
7
8   <groupId>com.library</groupId>
9   <artifactId>libraryManagement</artifactId>
10  <version>1.0-SNAPSHOT</version>
11  <packaging>jar</packaging>
12
13  <properties>
14    <!-- Java Version -->
15    <maven.compiler.source>1.8</maven.compiler.source>
16    <maven.compiler.target>1.8</maven.compiler.target>
17  </properties>
18
19  <dependencies>
20
21    <!-- Spring Context -->
22    <dependency>
23      <groupId>org.springframework</groupId>
24      <artifactId>spring-context</artifactId>
25      <version>5.3.39</version>
26    </dependency>
27
28    <!-- Spring AOP -->
29    <dependency>
30      <groupId>org.springframework</groupId>
31      <artifactId>spring-aop</artifactId>
32      <version>5.3.39</version>
33    </dependency>
34
35    <!-- Spring Web MVC -->
36    <dependency>
37      <groupId>org.springframework</groupId>
38      <artifactId>spring-webmvc</artifactId>
39      <version>5.3.39</version>
40    </dependency>
41  </dependencies>
42
43  <build>
44    <plugins>
45
46      <!-- Maven Compiler Plugin -->
47      <plugin>
48        <groupId>org.apache.maven.plugins</groupId>
49        <artifactId>maven-compiler-plugin</artifactId>
50        <version>3.11.0</version>
51
52        <configuration>
53          <source>1.8</source>
54          <target>1.8</target>
55        </configuration>
56      </plugin>
57
58      <!-- Exec Plugin -->
59      <plugin>
60        <groupId>org.codehaus.mojo</groupId>
61        <artifactId>exec-maven-plugin</artifactId>
62        <version>3.1.0</version>
63      </plugin>
64    </plugins>
65  </build>
66
67 </project>
```

```
[INFO] Nothing to compile - all classes are up to date
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.948 s
[INFO] Finished at: 2026-06-30T00:14:12+05:30
[INFO] -----
```